

The potential set of the light cone

A remark for the Mathematical Theory Atlas

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Status. Definitions and Propositions 2–3 are proved here from the explicit formula and Weil’s criterion; the converse in Proposition 2(iii) is Krein’s extension theorem and is cited; Computation 4 is taken from [7, Table 1]; Remarks make no claim.

Setup

Let W be the Weil distribution on $\mathbb{R}$, so that for every even $g \in C_c^\infty(\mathbb{R})$ the explicit formula reads

$$\sum_{\rho} \widehat{g}(\gamma_{\rho}) = \langle W, g \rangle, \quad \langle W, g \rangle = \widehat{g}\left(\frac{i}{2}\right) + \widehat{g}\left(-\frac{i}{2}\right) - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (g(\log n) + g(-\log n)) + W_{\infty}(g), \quad (1)$$

where $\rho = \frac{1}{2} + i\gamma_{\rho}$ runs over the nontrivial zeros with multiplicity, $\gamma_{\rho} \in \mathbb{C}$, and W_{∞} is the archimedean term [3, 7]. For g supported in $(-2a, 2a)$ only the prime powers with $\log n < 2a$ enter: the light cone of [7, §12]. Write $T^*(a) = 2\pi e^{2a}$ for its horizon.

Definition 1 (The potential set). For $a > 0$ let $\mathcal{P}(a)$ be the set of positive Radon measures ν on $\mathbb{R}$, symmetric under $t \mapsto -t$ and with $\int (1 + t^2)^{-1} d\nu < \infty$, such that

$$\int_{\mathbb{R}} \widehat{g} d\nu = \langle W, g \rangle \quad \text{for every even } g \in C_c^\infty(-2a, 2a). \quad (2)$$

An element of $\mathcal{P}(a)$ is a configuration of real “zeros” that no prime power inside the cone can distinguish from the true zeros. Put $\nu_{\zeta} = \sum_{\rho} \delta_{\gamma_{\rho}}$ when all γ_{ρ} are real.

Proposition 2. (i) $\mathcal{P}(a)$ is nonincreasing in a : $a < a'$ implies $\mathcal{P}(a') \subseteq \mathcal{P}(a)$.

(ii) If $\mathcal{P}(a) \neq \emptyset$ then the Weil form is nonnegative at support a : $\langle W, f * \tilde{f} \rangle \geq 0$ for every $f \in C_c^\infty(-a, a)$, in both the even and the odd sector.

(iii) Conversely, if $\langle W, f * \tilde{f} \rangle \geq 0$ for every $f \in C_c^\infty(-a, a)$, then $\mathcal{P}(a) \neq \emptyset$ (Krein’s extension theorem [5], in its distributional form [6]).

Proof. (i) The condition (2) for a' contains the condition for a . (ii) For $f \in C_c^\infty(-a, a)$ the function $g = f * \tilde{f}$ is even, supported in $(-2a, 2a)$, and $\widehat{g} = |\widehat{f}|^2 \geq 0$; hence $\langle W, g \rangle = \int |\widehat{f}|^2 d\nu \geq 0$ for any $\nu \in \mathcal{P}(a)$. (iii) The restriction of W to $(-2a, 2a)$ is a positive-definite distribution. Krein’s theorem produces a positive-definite distribution on $\mathbb{R}$ agreeing with it on $(-2a, 2a)$; by the Bochner–Schwartz theorem that distribution is the Fourier transform of a positive tempered measure ν , which is symmetric because W is even, and (2) holds by construction.¹ $\square$

¹Krein’s original statement concerns continuous positive-definite functions on an interval. The operator-theoretic proof (self-adjoint extensions of the generator of translations on the Hilbert space of the form) uses only the Hilbert-space structure and translation invariance and applies to positive-definite distributions; see [6]. The reader who prefers not to rely on this may take (iii) as the definition of the regime in which $\mathcal{P}(a)$ is studied.

Proposition 3 (The limit of the potential set).

$$\bigcap_{a>0} \mathcal{P}(a) = \begin{cases} \{\nu_\zeta\} & \text{if the Riemann Hypothesis holds,} \\ \emptyset & \text{otherwise.} \end{cases}$$

Proof. Let $\nu \in \bigcap_a \mathcal{P}(a)$. Then (2) holds for every even $g \in C_c^\infty(\mathbb{R})$, and by (1)

$$\int_{\mathbb{R}} \hat{g} d\nu = \sum_{\rho} \hat{g}(\gamma_{\rho}) \quad \text{for all even } g \in C_c^\infty(\mathbb{R}). \quad (3)$$

If all γ_{ρ} are real, the right side is $\int \hat{g} d\nu_{\zeta}$, and since the functions $\hat{g}$ are dense in the Schwartz space, the two tempered measures ν and ν_{ζ} coincide; conversely $\nu_{\zeta} \in \mathcal{P}(a)$ for every a by (1). If some γ_{ρ_0} is not real, Weil’s criterion [3, 4] provides $f \in C_c^\infty(\mathbb{R})$ such that, for $g = f * \tilde{f}$, $\sum_{\rho} \hat{g}(\gamma_{\rho}) < 0$; then (3) would give $\int |\hat{f}|^2 d\nu < 0$, impossible for a positive measure. So the intersection is empty. $\square$

Computation 4 (The identity phase). For $a < \frac{1}{2} \log 2 = 0.346574$ no prime power lies inside the cone, and the Weil form at support a is the archimedean and polar part alone. In the odd sector this prime-free form is positive definite for $a < 0.371601$ and indefinite beyond [7, Table 1, row $n = 2$]. Hence, in the odd sector, $\mathcal{P}(a)$ is nonempty for $a < 0.3716$ on the strength of the archimedean term alone, and the first prime power required to keep it nonempty is 2, which enters at $a = 0.3466$ and is needed by $a = 0.3716$. The corresponding entry points and deadlines for 3, 4, 5, 7, 8, 9 are the remaining rows of the same table.

Remark 5 (What the identity fixes and what the primes fix). As $a \downarrow 0$ the test functions in (2) concentrate near $x = 0$, where W is dominated by the singularity of W_{∞} ; this constrains only the large-scale density of ν , which must be that of the Riemann–von Mangoldt law $\frac{1}{2\pi} \log \frac{t}{2\pi}$. Every prime power that enters the cone adds a constraint at the frequency $\log n$, that is, on the fluctuations of ν at the scale $2\pi/\log n$. In this sense the identity prescribes how many zeros there are and the primes prescribe where they are, one frequency at a time, and $\mathcal{P}(a)$ is what remains undetermined at support a .

Remark 6 (The control). Nothing in Proposition 2 uses the Euler product. For the Davenport–Heilbronn function the same construction gives a family $\mathcal{P}_{\text{DH}}(a)$, nonempty for small a by the same archimedean mechanism, and by [7, Remark 12.9] expected to become empty near $a \approx 2.1$, where the horizon $\frac{2\pi}{5} e^{2a}$ reaches the off-line zero at height 85.7. The potential set can therefore contain configurations that are never realized by any zeta function; that it collapses to ν_{ζ} for ζ itself is the Riemann Hypothesis, not a property of the construction.

Remark 7 (Dictionary). The symbolic line “ $1 = A\Omega$ ” reads: at $a = 0^+$ the potential set is everything with the right density; “the first distinction is 2” is Computation 4; “potential defined by the primes” is Definition 1; and the subscript α, ω is the limit $a \rightarrow \infty$ of Proposition 3. The line is not an equation and should not be presented as one; the definition is.

Second remark: the ground-state transform and the spectral form of positivity

Status. This section re-derives, from the explicit formula alone, the identity that the external review of 29 September 2026 attributes to the source of the atlas entries T51–T53. Proposition 8 and Corollary 9 are proved here; the facts about the theta kernel are classical and cited. The inequality in Corollary 9 is not proved.

The theta kernel

Let ϕ be the even function on $\mathbb{R}$ whose Fourier transform is Riemann's Ξ : $\widehat{\phi}(z) = \int \phi(u) e^{izu} du = \Xi(z) = \xi(\frac{1}{2} + iz)$. Explicitly $\phi(u) = \sum_{n \geq 1} (2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}) \exp(-\pi n^2 e^{2u})$ for $u \geq 0$ [1, §10.1]. Three facts are used: $\phi > 0$ on $\mathbb{R}$ (for $u \geq 0$ each term is positive since $2\pi n^2 e^{2u} > 3$, and ϕ is even); ϕ and all its derivatives decay like $\exp(-\pi e^{2|u|})$; and $\widehat{\phi}(\pm i/2) = \xi(0) = \xi(1) = \frac{1}{2}$. Put

$$d\nu(u) = 2 \cosh(u/2) \phi(u) du, \quad \nu(1) = 2 \widehat{\phi}(i/2) = 1,$$

a probability measure, and for h in $L^2(\nu)$ write $\nu(h) = \int h d\nu$, $\text{Var}_\nu(h) = \int |h|^2 d\nu - |\nu(h)|^2$, and $b(h) = \int \tanh(u/2) h d\nu = \int 2 \sinh(u/2) \phi h du$.

Null directions

For $j \geq 0$ the Fourier transform of $\phi^{(j)}$ is $(-iz)^j \Xi(z)$, which vanishes at every zero of Ξ , on or off the critical line. Since $\phi^{(j)}$ and any $q \in C_c^\infty(\mathbb{R})$ satisfy the hypotheses of the explicit formula, the completed Weil form obeys

$$Q(\phi^{(j)}, q) = \sum_\rho \overline{\widehat{\phi^{(j)}}(\gamma_\rho)} \widehat{q}(\gamma_\rho) = 0 \quad (j \geq 0, q \in C_c^\infty(\mathbb{R}) \text{ or } q = \phi^{(k)}, k \geq 0), \quad (4)$$

where $Q(f, q) = \langle W, f * \tilde{q} \rangle$ is the sesquilinear form of the Weil distribution. In particular ($j = 0$) $W * \phi = 0$ pointwise, which in the x -space form of the archimedean term reads

$$c_1 \phi(u) + \int_{\mathbb{R}} (\phi(u) - \phi(v)) J_\Gamma(u - v) dv + \cosh(u/2) = \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (\phi(u + \log n) + \phi(u - \log n)), \quad (5)$$

with $J_\Gamma(s) = e^{-|s|/2} / (1 - e^{-2|s|})$ and a constant c_1 (namely $-(\gamma + \log \pi) + \int_{\mathbb{R}} (e^{-2|s|} - e^{-|s|/2})(1 - e^{-2|s|})^{-1} ds$); the term $\cosh(u/2)$ is the polar kernel $2 \cosh((u-v)/2)$ applied to ϕ , using $\int \cosh(v/2) \phi = \frac{1}{2}$ and $\int \sinh(v/2) \phi = 0$.

Proposition 8 (Ground-state transform). *For $h \in C_c^\infty(\mathbb{R})$, with $f = \phi h$,*

$$Q(f) = E_\Gamma(h) + E_P(h) - \frac{1}{2} \text{Var}_\nu(h) - \frac{1}{2} |b(h)|^2, \quad (6)$$

where

$$E_\Gamma(h) = \frac{1}{2} \iint J_\Gamma(u-v) \phi(u) \phi(v) |h(u) - h(v)|^2 du dv, \quad E_P(h) = \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \int \phi(u) \phi(u + \log n) |h(u + \log n) - h(u)|^2 du$$

Proof. Write $Q(f) = A(f) + P_r(f) + P_o(f)$ with A the archimedean part, P_r the prime part and P_o the polar part. In x -space, for $g = f * \tilde{f}$, $A(f) = c_1 \|f\|^2 + \int (g(0) - g(s)) J_\Gamma(s) ds = c_1 \|f\|^2 + \frac{1}{2} \iint J_\Gamma(u-v) |f(u) - f(v)|^2$, the last step by expanding the square; the singularity $J_\Gamma(s) \sim 1/(2|s|)$ is integrable against $|f(u) - f(v)|^2 = O(|u-v|^2)$. Next, $P_r(f) = -\sum_n \frac{\Lambda(n)}{\sqrt{n}} (g(\log n) + g(-\log n)) = -2 \sum_n \frac{\Lambda(n)}{\sqrt{n}} \text{Re} \int \overline{f(u)} f(u - \log n) du$. Expanding the square in E_P and substituting $u \mapsto u + \log n$ in one term,

$$E_P(h) = \int \phi(u) |h(u)|^2 \sum_n \frac{\Lambda(n)}{\sqrt{n}} (\phi(u + \log n) + \phi(u - \log n)) du + P_r(f).$$

Insert (5) for the inner sum: $E_P(h) = c_1 \|f\|^2 + \int \phi(u) |h(u)|^2 \int (\phi(u) - \phi(v)) J_\Gamma(u-v) dv du + \int \cosh(u/2) \phi |h|^2 + P_r(f)$. Hence

$$A(f) + P_r(f) = E_P(h) + \frac{1}{2} \iint J_\Gamma |\phi h(u) - \phi h(v)|^2 - \int \phi(u) |h(u)|^2 \int (\phi(u) - \phi(v)) J_\Gamma(u-v) dv du - \frac{1}{2} \int |h|^2 dv,$$

and the two J_Γ terms combine, after symmetrizing in u, v , to $\frac{1}{2} \iint J_\Gamma \phi(u) \phi(v) |h(u) - h(v)|^2 = E_\Gamma(h)$. Finally $\cosh((u-v)/2) = \cosh(u/2) \cosh(v/2) - \sinh(u/2) \sinh(v/2)$ gives $P_o(f) = 2 \int \cosh(u/2) \phi |h|^2 - 2 \int \sinh(u/2) \phi |h|^2 = \frac{1}{2} |\nu(h)|^2 - \frac{1}{2} |b(h)|^2$. Adding the pieces gives (6). All interchanges are justified by the superexponential decay of ϕ and the compact support of h . $\square$

Put $D(h, k) = 2(E_\Gamma + E_P)(h, k) - \text{Cov}_\nu(h, k)$, so that $Q(\phi h, \phi k) = \frac{1}{2} [D(h, k) - \overline{b(h)} b(k)]$, and let $h_* = \phi' / \phi$. Then $\nu(h_*) = \int 2 \cosh(u/2) \phi' = 0$ and $b(h_*) = \int 2 \sinh(u/2) \phi' = -\int \cosh(u/2) \phi = -\frac{1}{2}$ by parts, and (4) with $j = 1$ gives $D(h_*, h) = \overline{b(h_*)} b(h) = -\frac{1}{2} b(h)$ and $D(h_*, h_*) = \frac{1}{4}$. Let $\mathcal{H}$ be the span of $C_c^\infty(\mathbb{R})$ and h_* (the form (6) extends to it by the decay of ϕ), and define $\mathcal{P}h = h + 2b(h)h_*$.

Corollary 9 (Spectral form of Weil positivity). $b(\mathcal{P}h) = 0$, $D(h, h) = D(\mathcal{P}h, \mathcal{P}h) + |b(h)|^2$, and $Q(\phi h) = \frac{1}{2} D(\mathcal{P}h, \mathcal{P}h)$ for $h \in \mathcal{H}$. Consequently the Riemann Hypothesis is equivalent to

$$E_\Gamma(h) + E_P(h) \geq \frac{1}{2} \text{Var}_\nu(h) \quad \text{for all } h \in \mathcal{H}. \quad (7)$$

Moreover, if (7) holds, equality is attained on the infinite-dimensional space spanned by the centered functions $\phi^{(j)} / \phi - \nu(\phi^{(j)} / \phi)$, $j \geq 2$.

Proof. $b(\mathcal{P}h) = b(h) + 2b(h)b(h_*) = 0$. Expanding $D(h + ch_*, h + ch_*)$ with $c = 2b(h)$ and the two relations above gives $D(h, h) - 2|b(h)|^2 + |b(h)|^2$. The third identity is (6) rewritten. If (7) holds, $D \geq 0$ on $\mathcal{H}$ and $Q(\phi h) \geq 0$ for every $h \in C_c^\infty$; since $\phi > 0$, every $f \in C_c^\infty$ is of the form ϕh , and Weil's criterion gives RH. Conversely, under RH $Q \geq 0$ on C_c^∞ , and $Q(\phi \mathcal{P}h) = Q(\phi h + 2b(h)\phi') = Q(\phi h)$ by (4), so $D(\mathcal{P}h, \mathcal{P}h) \geq 0$ for $h \in C_c^\infty$, whence $D(h, h) \geq |b(h)|^2 \geq 0$; on $\mathcal{H}$ the same follows by the Cauchy–Schwarz inequality in the nonnegative form D applied to h and h_* . For the last claim, (4) for $\phi^{(j)}$ gives $D(\phi^{(j)} / \phi, h) = \overline{b(\phi^{(j)} / \phi)} b(h)$ for all h ; integrating by parts, $b(\phi^{(2k)} / \phi) = 0$ and $\nu(\phi^{(2k+1)} / \phi) = 0$, so the centered even-order quotients are null vectors of D , and $D = 0$ on a vector means equality in (7). Linear independence holds because a finite relation would make a nonzero polynomial times Ξ vanish identically. $\square$

Remark 10. The form $E_\Gamma + E_P$ is the Dirichlet form of a reversible Markov generator L on $L^2(\nu)$: gamma couples nearby positions with conductance $J_\Gamma(u-v)\phi(u)\phi(v)$, and each prime power couples u with $u \pm \log n$ with conductance $\frac{\Lambda(n)}{\sqrt{n}} \phi(u)\phi(u \pm \log n)$. In that language (7) says that the bottom of the spectrum of L on mean-zero functions is exactly $\frac{1}{2}$, and Corollary 9 adds that $\frac{1}{2}$ is then an eigenvalue of infinite multiplicity. This is why no uniform improvement of the constant is possible, and it is the same fact as the rigidity of [2, §2.4]: positivity of the Weil form has no slack. The atlas entries T51–T53 and the review's inequality IR-7 are this corollary. What is missing is a proof of (7); the identity (6) only relocates the difficulty from the sign of Q to a spectral gap, in a form where the equality directions are known.

References

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